

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I Bhavya PS (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CLASSICAL CRYPTOGRAPHY – SCENARIO BASED ASSESSMENT

Course: Cryptography and Network Security

Assessment: Classical Cryptography

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Scenario 1: Event Communication Security

Problem Statement

A student organization wants to securely share confidential event details with its core members using a classical encryption method. The original message is:

"MEET AT NOON"

A Caesar Cipher with a shift of 4 is used for encryption.

Aim

To encrypt and decrypt a message using the Caesar Cipher with a shift value of 4.

Algorithm Used

Caesar Cipher

- Each letter is shifted by a fixed number of positions.
- Here, every alphabet is shifted 4 positions forward.
- Spaces remain unchanged.

Encryption Process

Shift = +4

Original Letter	Shift (+4)	Encrypted Letter
M	Q	Q
E	I	I
E	I	I
T	X	X
A	E	E
T	X	X
N	R	R
O	S	S
O	S	S
N	R	R

Final Encrypted Message :-

MEET AT NOON

↓

QIIX EX RSSR

Answer (1a)

The encrypted message after applying a shift of 4 is

QIIX EX RSSR

Decryption Process

The authorized member knows that the message was encrypted using a shift of 4.

Therefore, every letter is shifted 4 positions backward.

Example:

Q → M

I → E

I → E

X → T

E → A

X → T

R → N

S → O

S → O

R → N

Thus,

QIIX EX RSSR

↓

MEET AT NOON

Answer (1b)

The authorized member decrypts the message by shifting every encrypted letter 4 positions backward, restoring the original message.

Result

Original Message:

MEET AT NOON

Encrypted Message:

QIIX EX RSSR

Recovered Message:

MEET AT NOON

Scenario 3: Secure Messaging System

Problem Statement

A company deploys a keyword-based encryption technique for secure internal communication. An employee wants to encrypt the message "HELLO" using the keyword "KEY".

Aim

To encrypt a message using the Vigenère Cipher with the keyword KEY.

Algorithm Used

Vigenère Cipher

- The plaintext is encrypted using a repeating keyword.
- Each letter of the keyword determines how many positions the corresponding plaintext letter is shifted.
- If the keyword is shorter than the message, it is repeated until it matches the message length.

Step 1: Repeat the Keyword

Plaintext: H E L L O

Keyword: K E Y K E

(The keyword KEY is repeated to become KEYKE.)

Step 2: Convert Letters to Numbers

Using A = 0, B = 1, C = 2, ..., Z = 25

Plaintext	H	E	L	L	O
Value	7	4	11	11	14
Keyword	K	E	Y	K	E
Value	10	4	24	10	4

Step 3: Encryption Formula

Cipher Value = (Plaintext Value + Keyword Value) mod 26

Plaintext	Value	Keyword	Value	Calculation	Cipher Value	Cipher Letter
H	7	K	10	$(7 + 10) \bmod 26 = 17$	17	R
E	4	E	4	$(4 + 4) \bmod 26 = 8$	8	I
L	11	Y	24	$(11 + 24) \bmod 26 = 9$	9	J
L	11	K	10	$(11 + 10) \bmod 26 = 21$	21	V

O	14	E	4	$(14 + 4) \bmod 26 = 18$	18	S
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Encrypted Message :-

HELLO

↓

RIJVS

Answer (3a)

The plaintext HELLO is encrypted using the keyword KEY. The keyword is repeated as KEYKE, and each plaintext letter is shifted according to the corresponding keyword letter. The final encrypted message is:

RIJVS

Answer (3b)

The keyword KEY is shorter than the plaintext, so it is repeated (KEYKE) until it matches the length of the message. Each letter of the keyword provides a different shift value:

- K shifts the first letter by 10 positions.
- E shifts the second letter by 4 positions.
- Y shifts the third letter by 24 positions.
- The keyword repeats, so K and E are used again for the fourth and fifth letters.

Because the shift changes for each character instead of remaining constant, the Vigenère Cipher is more secure than the Caesar Cipher and makes frequency analysis more difficult.

Result

- Plaintext: HELLO
- Keyword: KEY
- Repeated Keyword: KEYKE
- Ciphertext: RIJVS

Conclusion: The Vigenère Cipher successfully encrypts the message by using a repeating keyword, where each character is encrypted with a different shift value determined by the corresponding keyword letter. This provides stronger security than a fixed-shift cipher like the Caesar Cipher.